Corrected and conduct of chapter 8

Use as a teacher. The following answers relate to the outstanding tasks in the course. first clarify the approach; provide graduated help before the solution. estimates to be calibrated in class.

Diagnosis and Remediation

1. 4. *Action:* Add and divide by the number of values.
2. Yes: positive sum numbers 1. *Action:* Put at the same denominator.
3. 9. *Action:* Keep the parentheses before the square.
4. 0,4. *Action:* Use the supplement.
5. 5. *Action:* Write net gain = received less paid.
6. 0,5. *Action:* Check the square of the answer.

Research activity

Gains (-1; 0; 1) and (-3; 0; 3): expectation 0 in both cases, variances $\frac{2}{3}$ and 6. The second is three times more dispersed in standard deviation.

Immediate applications

1. Values 0.1.2; probabilities $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. $P(X \leq 1) = \frac{3}{4}$.
2. $E(G) = \frac{10}{5}$ = 2. For a bet m, $E(G - m) = 2 - m$; Fairness 2.
3. $E(X) = 0$, $E(X^2)$ = 1, variance 1 and standard deviation 1.
4. $E(Y) = -1$, $V(Y) = 16$, $\sigma(Y) = 4$.
5. $\frac{2\sigma}{\sqrt{n}}$ = 1; interval [4; 6].

Exercises: approach and results

Fiscal year 8.1 · A law to be built. Values 0.1.2; each obtained by two sides, probabilities $\frac{1}{3}$. $P(X < 2) = \frac{2}{3}$. Attention to the rest 0 for 3 and 6.

Exercise 8.2 · Two variance calculations. $E(X) = -\frac{1}{2} + 1 = \frac{1}{2}$. $E(X^2) = 1 + 4 = 5$; $V = 5 - \frac{1}{4} = \frac{19}{4}$, $\sigma = \frac{\sqrt{19}}{2}$. Direct: $(25\frac{}{4})(\frac{1}{4}) + (\frac{1}{4})(\frac{1}{2}) + (\frac{49}{4})(\frac{1}{4}) = \frac{76}{16} = \frac{19}{4}$.

Exercise 8.3 · Fair play. $E(G) = \frac{15}{5}$ = 3, so m=3. $E(G^2) = \frac{125}{5} = 25$; $V(G) = 25 - 9$ = 16, standard deviation 4, unchanged by bet.

Exercise 8.4 · Same average, other risk. $E(X) = E(Y) = 1$; $V(X) = 0$; $E(Y^2) = 10$, $V(Y)$ = 9. X is certain, Y dispersed. One cannot classify a player's preferences only from expectation.

Exercise 8.5 · Linearity without independence. $E(X) = \frac{1+2+3+4+5+6}{6} = \frac{7}{2}$; $E(Y) = 7 - E(X) = \frac{7}{2}$. Like $X + Y = 7$, his expectation is equal to 7, sum of the two expectations. $\{X = 1\}$ and $\{Y = 6\}$ are identical: their intersection has probability $\frac{1}{6}$, while the product of their probabilities is $\frac{1}{36}$. The variables are not independent. Linearity remains valid without this hypothesis.

Exercise 8.6 · Minimum loss. Write $X - a = (X - \mu) + (\mu - a)$, develop and then use $E(X - \mu)$ = 0. Square ($a - \mu)^2$ is minimal, zero only for $a = \mu$. The average minimizes Average quadratic deviation.

Exercise 8.7 · Frequency of letters. Pass text in tiny, keep characters for which `isalpha()` is true, then count "a". Divide this number by the number of letters if for a text without a letter, return it `None`, because the frequency is not defined. `Aba !`, two "a" out of three letters give $\frac{2}{3}$; the other two tests give `None`. The programme sent `frequence_a` follows this agreement.

Exercise 8.8 · Deferred review. Law: 0 , 1, 2 probability $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$. Expectation 1 ; time Order 2: $\frac{1}{2} + 1 = \frac{3}{2}$, variance $1\frac{-}{2}$. The four paths are explicitly counted.

CQFD, Exercise 28 p. 306, all questions. $E(X) = (-10) \times 0{,}35 + (-5) \times 0{,}15 + 0 \times 0{,}4 + 20 \times 0{,}1 = -2{,}25 = -\frac{9}{4}$. A negative expectation is compatible with a possible positive value.

CQFD, exercise 31 p. 306, all questions; use the law of fiscal year 28. $E(X^2) = 100 \times 0{,}35 + 25 \times 0{,}15 + 400 \times 0{,}1 = 78{,}75$. Thus $V(X) = 78{,}75 - (-2{,}25)^2 = 73{,}6875 = \frac{1179}{16}$then $\sigma(X) = \frac{\sqrt{1179}}{4} \approx 8{,}6$. Do not round the expectation before the variance calculation.

MCQ: answers and reasons

MCQ 1.

1. B. Partition of the universe.

2. A. $0 + \frac{4}{4}$.

3. D. König-Huygens.

4. C. Positive root.

5. B. Linearity.

MCQ 2.

1. C. Translation without new dispersion.

2. A. Absolute factor value.

3. D. A variable may be negative or have an unmet average.

4. B. Sufficient common finite universe.

5. C. $\frac{4}{10} = 0{,}4$.

True or false

1. False: this is an average of squares with positive weights.

2. False: constant 1 and equiprobable variable -2, 4 have the same average.

3. True: deviations from the new average remain the same.

Differentiation. For technical blocking, propose the first step only (domain, a reference factor or universe), and then remove the help. For an autonomous student, ask for a second method and treatment of a limited case. In oral, allow an initial reformulation in the language mastered, then require a precise mathematical sentence in French.